ModelIt!

Student Activity Pack

DNA Decoded: Why You Are (Mostly) Unique

The Blueprint of Life and What Makes You, You

Grade 07L2 | NGSS MS-LS3-1, MS-LS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think you look similar to your parents but not identical to either one?

2 What do you know about DNA and what it does in your body?

3 Draw what you think happens when traits from two parents combine in their offspring.

4 Why do you think identical twins look the same but siblings do not?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Parent Gene Variation | Environmental Conditions | Gene Expression Level | Trait Variation in Offspring | Survival Advantage

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: High Parent Variation

Diverse parent genes, stable environment

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Environment Changes

Same genes, shifting environmental conditions

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Low Variation (Asexual-like)

Minimal parent variation, changing environment

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Mutation Rate — The frequency at which random changes occur in DNA during cell division, creating new variations

Epigenetic Factors — Chemical modifications that change gene activity without altering the DNA sequence itself

Sexual Selection — When organisms choose mates based on certain traits, influencing which genes pass to the next generation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If you share 99.9% of your DNA with every other human, what makes you unique? Use evidence from your simulation to support your answers.

1 Why is genetic variation important for a species' long-term survival?

2 How does your model show the difference between sexual and asexual reproduction outcomes?

3 What would happen to a population if all individuals had identical DNA and the environment suddenly changed?

4 How do environmental conditions affect gene expression even when DNA stays the same?

5 Why might a trait that is harmful in one environment be beneficial in another?

8 STEM Challenge: Design a Genetic Diversity Exhibit

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YOUR ENGINEERING MISSION

Create an interactive exhibit demonstrating why genetic diversity is essential for species survival, using evidence from your simulation model.

THE SCENARIO

A science museum is opening a new genetics wing and needs an interactive exhibit that shows visitors why genetic diversity matters. Your team will design the exhibit using data from your model showing how variation affects survival.

Engineering Design Thinking

1. WHAT INTERACTIVE ELEMENTS WILL HELP VISITORS UNDERSTAND GENE EXPRESSION?

2. HOW WILL YOU SHOW THAT ENVIRONMENT INFLUENCES WHICH TRAITS ARE ADVANTAGEOUS?

3. WHAT MODEL DATA WILL YOU DISPLAY TO PROVE GENETIC DIVERSITY MATTERS?

4. HOW WILL YOU REPRESENT THE DIFFERENCE BETWEEN SEXUAL AND ASEXUAL REPRODUCTION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Genetic Counselors help individuals and families understand genetic conditions, inheritance patterns, and testing options. They work in hospitals, clinics, and research labs, earning $80,000-$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 A farmer plants a field with genetically identical crop plants (produced by asexual reproduction). A new fungal disease arrives and destroys the entire crop. Based on the genetics model, which explanation best accounts for why the entire field was lost?

- **A.** Asexual reproduction produces weaker organisms that are always more susceptible to disease
- **B.** All plants had identical DNA, so if one lacked resistance to the fungus, none of them had resistance
- **C.** The fungal disease was too strong for any plant to survive regardless of genetic variation
- **D.** Asexual reproduction causes more harmful mutations than sexual reproduction

2 Two siblings share the same parents but look noticeably different from each other. According to the model, which factor is most responsible for their physical differences?

- **A.** One sibling received more DNA from the mother and the other received more from the father
- **B.** Sexual reproduction randomly combines different allele combinations from each parent during meiosis, creating unique offspring every time
- **C.** Siblings always have completely different genes because each parent produces entirely new DNA for each child
- **D.** Physical differences between siblings are caused entirely by environmental conditions, not genetics

3 In the model, when Environmental Conditions change but Parent Gene Variation stays constant, some offspring gain a Survival Advantage while others lose it. Which crosscutting concept best explains this observation?

- **A.** Structure and Function: the structure of DNA determines the function of every organism
- **B.** Cause and Effect: the environmental change causes certain trait combinations to become advantageous or disadvantageous
- **C.** Scale and Proportion: larger organisms always have better survival advantages
- **D.** Stability and Change: organisms always maintain stability regardless of environmental conditions

4 A student argues that a mutation that causes sickle-shaped red blood cells is always harmful. Using the model's concept of environmental influence on survival advantage, which response best challenges this claim?

- **A.** All mutations are neutral and have no effect on survival
- **B.** In regions with malaria, carrying one copy of the sickle cell allele provides resistance to the disease, making the same mutation beneficial in that environmental context
- **C.** Sickle cell mutations are beneficial everywhere because they make blood cells stronger
- **D.** Mutations cannot be both harmful and beneficial because DNA is either functional or broken

5 **In the model, when Parent Gene Variation is set to low (simulating asexual-like reproduction), Trait Variation in Offspring is minimal and Survival Advantage drops sharply when Environmental Conditions change. A student concludes that asexual reproduction is always inferior to sexual reproduction. Why is this conclusion too broad?**

A. Asexual reproduction is actually always superior because it produces more offspring faster

B. Asexual reproduction is highly efficient in stable environments because identical offspring are already well-adapted, but it becomes dangerous when conditions change

C. The model is inaccurate because asexual organisms never face environmental changes

D. Asexual reproduction produces more genetic variation than sexual reproduction through faster mutation rates